

Multifile LaTeX Document Example

Your Name

May 13, 2026

Contents

0.2 Notation

0.3 Weak form

0.4 Notes

Topics to look into:

- Hilbert spaces
- Sobolev spaces
- H1 space
- Square integrable functions
- poincare inequality - with reference to bounds on functions using bounds on its deriv and geometric constraints on the domain

0.4.1 Notation

- $H^1(0, 1)$ is the space of all functions u such that $u \in L^2(0, 1)$ and $u' \in L^2(0, 1)$.
- $H_0^1(0, 1)$ is the space of all functions u such that $u \in H^1(0, 1)$ and $u(0) = u(1) = 0$.
- Dirichlet boundary conditions: $u(0) = \alpha$ and $u(1) = \beta$
- Neumann boundary conditions: $u'(0) = \alpha$ and $u'(1) = \beta$
- Mixed boundary conditions: $u(0) = \alpha$ and $u'(1) = \beta$

In finite element methods and weak formulations, the notation

$$\|v\|_0$$

usually means the L^2 norm of the function v . It does *not* mean the “zero norm” from sparse optimization.

The subscript 0 means that we are measuring the function itself, with zero derivatives included.

$$\|v\|_0 = \|v\|_{L^2(a,b)} = \left(\int_a^b |v(x)|^2 dx \right)^{1/2}.$$

Therefore,

$$\|v'\|_0 = \left(\int_a^b |v'(x)|^2 dx \right)^{1/2}.$$

So $\|v\|_0$ measures the overall size, or magnitude, of the function v on the interval $[a, b]$. Equivalently,

$$\|v\|_0^2 = \int_a^b |v(x)|^2 dx,$$

which is the area under the curve $|v(x)|^2$.

Connection to Sobolev Norms

The notation $\|\cdot\|_0$, $\|\cdot\|_1$, $\|\cdot\|_2$, etc. is common in Sobolev spaces.

The zeroth-order Sobolev norm is just the L^2 norm:

$$\|v\|_0 = \|v\|_{L^2(a,b)}.$$

The first-order Sobolev norm is

$$\|v\|_1 = \|v\|_{H^1(a,b)} = (\|v\|_0^2 + \|v'\|_0^2)^{1/2}.$$

Written out fully,

$$\|v\|_1 = \left(\int_a^b |v(x)|^2 dx + \int_a^b |v'(x)|^2 dx \right)^{1/2}.$$

The second-order Sobolev norm is

$$\|v\|_2 = \|v\|_{H^2(a,b)} = (\|v\|_0^2 + \|v'\|_0^2 + \|v''\|_0^2)^{1/2}.$$

Written out fully,

$$\|v\|_2 = \left(\int_a^b |v(x)|^2 dx + \int_a^b |v'(x)|^2 dx + \int_a^b |v''(x)|^2 dx \right)^{1/2}.$$

General Pattern

For Sobolev spaces, the subscript on the norm tells us how many derivatives are included.

$$\|v\|_0 \quad \text{measures} \quad v$$

$$\|v\|_1 \quad \text{measures} \quad v \text{ and } v'$$

$$\|v\|_2 \quad \text{measures} \quad v, v', \text{ and } v''.$$

More generally,

$$\|v\|_{H^k(a,b)} = \left(\sum_{j=0}^k \|v^{(j)}\|_{L^2(a,b)}^2 \right)^{1/2}.$$

Using the shorthand notation,

$$\|v\|_k = \left(\sum_{j=0}^k \|v^{(j)}\|_0^2 \right)^{1/2}.$$

0.4.2 General Notes

Taken from *Numerical Mathematics by Jeffrey Ovall* Function spaces (vector spaces of functions) on the interval (a, b) :

- $L^2(a, b)$: Consists of all measurable functions on (a, b) that are square integrable. i.e. $\nu \in L^2(a, b) \Rightarrow \int_a^b |\nu(x)|^2 dx < \infty$
- For $\nu \in L^2(0, 1)$, we define the norm by: $\|\nu\|_0 = \sqrt{\int_0^1 \nu(x)^2 dx}$
- $H^1(a, b)$: Consists of all functions ν that are square integrable and have square integrable first derivatives.

$$H^1(a, b) = \{\nu \in L^2(a, b) : \nu' \in L^2(a, b)\}$$

- $H_0^1(a, b)$: Consists of all functions $\nu \in H^1(a, b)$ that vanish at the endpoints a and b .

$$H_0^1(a, b) = \{\nu \in H^1(a, b) : \nu(a) = \nu(b) = 0\}$$

- $H^2(a, b)$: Consists of all functions ν that are square integrable and have square integrable first and second derivatives.

$$H^2(a, b) = \{\nu \in L^2(a, b) : \nu'' \in L^2(a, b)\}$$

Linear and Bilinear Forms: Let V , W , and U be three vector spaces. Let $f : V \times W \rightarrow U$ be any map.

- The map f is linear in its first argument if, for every $w \in W$, the map $V \rightarrow U, v \mapsto f(v, w)$ is linear. This can be rewritten as, the map $f : V \times W \rightarrow U$ is linear in its first argument if and only if for every $\lambda_1, \lambda_2 \in \mathbb{R}$ and $v_1, v_2 \in V$ and $w \in W$, we have

$$f(\lambda_1 v_1 + \lambda_2 v_2, w) = \lambda_1 f(v_1, w) + \lambda_2 f(v_2, w)$$

- The map f is *linear* in its second argument if, for every $v \in V$, the map $W \rightarrow U, w \mapsto f(v, w)$ is linear. Similarly, the map $f : V \times W \rightarrow U$ is linear in its second argument if and only if for every $\lambda_1, \lambda_2 \in \mathbb{R}$ and $v \in V$ and $w_1, w_2 \in W$, we have

$$f(v, \lambda_1 w_1 + \lambda_2 w_2) = \lambda_1 f(v, w_1) + \lambda_2 f(v, w_2)$$

- The map f is *bilinear* if it is linear in both arguments.

Note: With reference to FEA, the L.H.S of the weak form is a bilinear form and the R.H.S is a linear form.

0.4.3 Introduction

from numerical mathematics by jeffrey ovall

General weak formulation

Given a function f and numbers α, β , find a function u such that

$$-u''(x) = f(x) \quad \text{in } (0, 1), \quad u(0) = \alpha, \quad u(1) = \beta \quad (1)$$

The *strong/classical* view of the model above requires that $u \in C^2(0, 1) \cap C[0, 1]$ satisfies the differential equation at each point in $(0, 1)$ as well as the boundary conditions. (from section 6.1 of textbook).

We can also view the problem in its weak form. Suppose that u satisfies the differential equation and that v is any sufficiently smooth function that satisfies $v(0) = v(1) = 0$. Then, we can integrate the differential equation over the domain $(0, 1)$ and use integration by parts to get

$$\begin{aligned} \int_0^1 f(x)v(x)dx &= - \int_0^1 u''(x)v(x)dx = \int_0^1 u'(x)v'(x)dx - [u'(x)v(x)]_{x=0}^{x=1} \\ &= \int_0^1 u'(x)v'(x)dx \end{aligned}$$

Note that the resulting equation no longer requires the second derivative of u . In fact, we can relax the smoothness assumptions on both u and f . The above integral equation is the basis for the weak (variational) form of the problem.

Definition 1: Let $f \in L^2(0, 1)$. The weak or variational form of (1) is as follows: Find $u \in H^1(0, 1)$ such that $u(0) = \alpha$ and $u(1) = \beta$ and

$$\int_0^1 u'(x)v'(x)dx = \int_0^1 f(x)v(x)dx \quad \forall v \in H_0^1(0, 1) \quad (2)$$

If we are interested in a problem with **mixed Dirichlet-Neumann boundary conditions**, the strong form of the mixed problem is

$$-u''(x) = f(x) \quad \text{in } (0, 1), \quad u(0) = \alpha, \quad u'(1) = \beta \quad (3)$$

The weak form of the mixed problem is:

$$\begin{aligned} \int_0^1 f(x)v(x)dx &= - \int_0^1 u''(x)v(x)dx = \int_0^1 u'(x)v'(x)dx - [u'(x)v(x)]_{x=0}^{x=1} \\ &= \int_0^1 u'(x)v'(x)dx - \beta v(1) \end{aligned}$$

Therefore, the weak form for this mixed BC problem is as follows: Find $u \in H^1(0, 1)$ such that $u(0) = \alpha$ and $u'(1) = \beta$ and

$$\int_0^1 u'(x)v'(x)dx = \int_0^1 f(x)v(x)dx + \beta v(1) \quad \forall v \in H_0^1(0, 1) \quad (4)$$

0.4.4 Integration by Parts and Boundary Terms in the Weak Form

Suppose we start with the strong form of Poisson's equation on a domain Ω :

$$-\Delta u = f \quad \text{in } \Omega.$$

To derive the weak form, we multiply both sides by a test function v and integrate over the domain:

$$\int_{\Omega} (-\Delta u) v \, dx = \int_{\Omega} f v \, dx.$$

The left-hand side contains second derivatives of u . The purpose of integration by parts is to move one derivative from u onto the test function v . This lowers the differentiability requirement on u , which is one of the main reasons weak forms are useful.

In multiple dimensions, integration by parts is based on the identity

$$\nabla \cdot (v \nabla u) = \nabla v \cdot \nabla u + v \Delta u.$$

Rearranging,

$$-v \Delta u = \nabla u \cdot \nabla v - \nabla \cdot (v \nabla u).$$

Therefore,

$$\int_{\Omega} (-\Delta u) v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} \nabla \cdot (v \nabla u) \, dx.$$

The first term is still an integral over the whole domain Ω :

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx.$$

The second term is converted into a boundary integral using the divergence theorem:

$$\int_{\Omega} \nabla \cdot (v \nabla u) \, dx = \int_{\partial\Omega} v \nabla u \cdot n \, ds,$$

where n is the outward unit normal vector on the boundary $\partial\Omega$. Since,

$$\nabla u \cdot n = \frac{\partial u}{\partial n},$$

we get

$$\int_{\Omega} \nabla \cdot (v \nabla u) \, dx = \int_{\partial\Omega} v \frac{\partial u}{\partial n} \, ds.$$

Thus,

$$\int_{\Omega} (-\Delta u) v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds.$$

Putting this back into the original equation gives the weak form with the boundary term kept:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds = \int_{\Omega} f v \, dx.$$

Integration by parts does not eliminate the domain integral. Instead, it transforms the expression into a new domain integral plus a boundary term:

$$\text{integration by parts} = \text{domain integral} + \text{boundary integral}.$$

In one dimension, this is the familiar formula

$$\int_a^b -u'' v \, dx = \int_a^b u' v' \, dx - [u' v]_a^b.$$

The term $[u' v]_a^b$ only depends on the endpoints a and b . In higher dimensions, the endpoints are replaced by the boundary $\partial\Omega$, so

$$[u' v]_a^b \longrightarrow \int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds.$$

Some boundary terms may disappear depending on the boundary conditions. For Dirichlet boundary conditions, we often build the boundary condition directly into the function space. For example, if

$$u = u_D \quad \text{on } \Gamma_D,$$

then the test functions are usually chosen to vanish on the Dirichlet boundary:

$$v = 0 \quad \text{on } \Gamma_D.$$

In that case, the boundary integral over Γ_D becomes zero because it contains the factor v :

$$\int_{\Gamma_D} \frac{\partial u}{\partial n} v \, ds = 0.$$

This is why, in the standard weak form with essential Dirichlet boundary conditions built into the space, the boundary term often disappears.

However, if we do **not** build the Dirichlet condition into the function space, then the test functions v do not necessarily vanish on Γ_D . In that case, the boundary term must be kept:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds = \int_{\Omega} f v \, dx.$$

This is the situation in methods where the Dirichlet condition is imposed by an additional equation rather than being built directly into the space.

0.4.5 Nonhomogeneous Dirichlet Data and Affine Spaces

For homogeneous Dirichlet boundary conditions,

$$u = 0 \quad \text{on } \Gamma_D,$$

the solution and test functions can both live in the space

$$V_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

This is a vector space: if $v_1, v_2 \in V_0$, then $v_1 + v_2 \in V_0$, and if $\alpha \in \mathbb{R}$, then $\alpha v_1 \in V_0$.

However, for nonhomogeneous Dirichlet boundary conditions,

$$u = u_D \quad \text{on } \Gamma_D,$$

where $u_D \neq 0$, the admissible set is

$$V_{u_D} = \{u \in H^1(\Omega) : u = u_D \text{ on } \Gamma_D\}.$$

This set is not a vector space. For example, if $u_1, u_2 \in V_{u_D}$, then on Γ_D ,

$$u_1 = u_D, \quad u_2 = u_D.$$

Therefore,

$$u_1 + u_2 = 2u_D \quad \text{on } \Gamma_D,$$

so in general $u_1 + u_2 \notin V_{u_D}$. Similarly, if $\alpha \neq 1$, then

$$\alpha u_1 = \alpha u_D \quad \text{on } \Gamma_D,$$

so $\alpha u_1 \notin V_{u_D}$ in general.

Thus, V_{u_D} is not a vector space. It is an **affine space**, meaning it is a shifted vector space. We write

$$V_{u_D} = u_D + V_0,$$

where u_D is one function satisfying the prescribed boundary data, and V_0 is the homogeneous Dirichlet space.

This motivates the standard finite element decomposition

$$u = u_0 + u_D,$$

where

$$u_D = \text{a known extension of the boundary data into } \Omega,$$

and

$$u_0 = \text{the unknown correction satisfying } u_0 = 0 \text{ on } \Gamma_D.$$

Indeed, on the Dirichlet boundary,

$$u_0 = u - u_D = u_D - u_D = 0.$$

Therefore,

$$u_0 \in V_0.$$

This is useful because V_0 is a true vector space, so the unknown correction u_0 can be represented using finite element basis functions with free coefficients. Substituting

$$u = u_0 + u_D$$

into the weak form gives

$$\int_{\Omega} \nabla(u_0 + u_D) \cdot \nabla v_0 \, dx = \int_{\Omega} f v_0 \, dx \quad \forall v_0 \in V_0.$$

Expanding the left-hand side,

$$\int_{\Omega} \nabla u_0 \cdot \nabla v_0 \, dx + \int_{\Omega} \nabla u_D \cdot \nabla v_0 \, dx = \int_{\Omega} f v_0 \, dx.$$

Since u_D is known, we move its contribution to the right-hand side:

$$\int_{\Omega} \nabla u_0 \cdot \nabla v_0 \, dx = \int_{\Omega} f v_0 \, dx - \int_{\Omega} \nabla u_D \cdot \nabla v_0 \, dx.$$

So the nonhomogeneous Dirichlet problem for u is converted into a homogeneous Dirichlet problem for u_0 . Once u_0 is computed, the full solution is recovered by

$$u = u_0 + u_D.$$

Summary:

- Homogeneous Dirichlet data gives a vector space of admissible functions.
- Nonhomogeneous Dirichlet data gives an affine space of admissible functions.
- An affine space is a shifted vector space.
- To use standard finite element machinery, we write $u = u_0 + u_D$.
- The unknown u_0 satisfies homogeneous Dirichlet conditions and therefore lives in a vector space.
- The known function u_D carries the prescribed nonzero boundary data.

A finite Element Method

0.5 NGSolve Notes

CoefficientFunctions: A `CoefficientFunction` is a function which can be evaluated on a mesh, and may be used to provide the coefficient or a right-hand-side to the variational formulation. Because typical finite element procedures iterate over elements, and map integration points from a reference element to a physical element, the evaluation of a `CoefficientFunction` requires a mapped integration point. The mapped integration point contains the coordinate, as well as the Jacobian, but also element number and region index. This allows the efficient evaluation of a wide class of functions needed for finite element computations.

TestFunction and TrialFunction: `TestFunction` and `TrialFunction` are symbolic objects used to define the variational (weak) form of a partial differential equation. They act as placeholders for the unknown solution and the weighting function during the construction of bilinear and linear forms, allowing you to write equations in a natural mathematical syntax. **Key Concepts and Usage**

- **Definition.** They are created from a defined space (`fes`), typically using `u = fes.TrialFunction()` and `v = fes.TestFunction()`, or simultaneously with `u, v = fes.TnT()`.
- **Symbolic operation.** They are used to symbolically express bilinear forms (e.g., $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx$) before assembly.
- **Spaces.** Test and trial functions belong to a `FESpace` (e.g. H^1 , L^2 , $H(\text{div})$) that defines their continuity and polynomial order.

GridFunction: A `GridFunction` is a function defined on a mesh that can be used to store and manipulate the solution of a partial differential equation. It is a container for the coefficient vector of a finite element space, and can be used to store the solution of a partial differential equation.

0.6 Matrix Decomposition

Let $A \in \mathbb{R}^{m \times n}$ be the matrix that solves the system described by $Ax = b$ for $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

After performing [Gaussian elimination](#) on A , U (row reduction of A) is in row echelon form if:

- All zero rows are at the bottom.
- The first nonzero entry in a row (from the left) is located to the right of the first nonzero entry in every row above.

The rank of A is the number of nonzero rows in U . We say the system $A\mathbf{x} = \mathbf{b}$ is inconsistent (i.e. no solution) if $\text{rank}(A) < \text{rank}([A\mathbf{b}])$. The system has a unique solution when $\text{rank}(A) = \text{rank}([A\mathbf{b}]) = n$ (i.e. $\text{rank}(A)$ equals the number of variables in the system). The system has infinitely many solutions when $\text{rank}(A) = \text{rank}([A\mathbf{b}])$ and $\text{rank}(A) < n$.

0.6.1 LU Decomposition

Any non-singular matrix A can be decomposed into a product of a lower triangular matrix L and an upper triangular matrix U such that $A = LU$, using the procedures of Gaussian elimination.

Theorem 1: If A can be reduced by Gaussian elimination to row echelon form using only operations of the form "add c times row j to row i " (that is, without scaling rows and without interchanging rows), then A has an LU decomposition of the form $A = LU$, where L is lower triangular and U is upper triangular.

In particular, after performing Gaussian elimination on A , the matrix U is the corresponding row echelon form of A , and L is given by

$$L = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ -c_{2,1} & 1 & 0 & \cdots & 0 \\ -c_{3,1} & -c_{3,2} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -c_{m,1} & -c_{m,2} & -c_{m,3} & \cdots & 1 \end{pmatrix}$$

where each entry corresponds to the elementary row operation "add $c_{i,j}$ times row j to row i " performed during Gaussian elimination.

If A can be decomposed into $A = LU$, then $\mathbf{x}$ can be solved for by:

1. Let $\mathbf{y} = U\mathbf{x}$ and solve the system for $L\mathbf{y} = \mathbf{b}$.
2. Once $\mathbf{y}$ is found, the upper triangular system $U\mathbf{x} = \mathbf{y}$ can be solved with back substitution for $\mathbf{x}$.

If we transform A into U by Gaussian elimination, the lower triangular matrix L can be found by performing negations of the same operations on the identity matrix in reverse order. There are other methods to find L and U , such as the Doolittle algorithm and the Crout algorithm.

Steps	
1.	Initialize L to an identity matrix, $\mathbb{I}$ of dimension $n \times n$, and $U = A$.
2.	For $i = 1, \dots, n$ do Step 3.
3.	For $j = i + 1, \dots, n$ do Steps 4–5.
4.	Set $l_{ji} = u_{ji}/u_{ii}$.
5.	Perform $U_j = (U_j - l_{ji}U_i)$ (where U_i, U_j represent the i and j rows of the matrix $\mathbf{U}$, respectively).

Once we have L and U , we can solve for as many right-hand side vectors $\mathbf{b}$ as desired very quickly using a two-step process. First, let $\mathbf{y} = U\mathbf{x}$ and then solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ by forward substitution.

Pseudocode for forward substitution

Steps	
1.	Set $y_1 = b_1/l_{11}$.
2.	For $i = 2, 3, \dots, n$ do Step 3.
3.	$y_i = \left(b_i - \sum_{j=1}^{i-1} l_{ij}y_j \right) / l_{ii}$

Then solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using backward substitution.

Pseudocode for backward substitution

We can also easily solve for A^{-1} and $\det(A)$ once L and U are known. For more details, see [Numerical Methods for Engineers](#).

Sources:

- <https://ubcmath.github.io/MATH307/systems/lu.html>
- https://johnfoster.pge.utexas.edu/numerical-methods-book/LinearAlgebra_LU.html

Steps	
1.	Set $x_n = y_n/u_{nn}$.
2.	For $i = n - 1, n - 2, \dots, 1$ do Step 3.
3.	$x_i = \left(y_i - \sum_{j=i+1}^n u_{ij}x_j \right) / u_{ii}$

0.7 Jacobi Method

- **Linear system:** $Ax = b$
- **Residual:** $r = b - Ax_k$
- **Error:** $e = x - x_k$
- **Preconditioner:** $P \approx A$

The Jacobi method is a simple iterative method for solving linear systems of equations. It is a first-order method, and is given by the following update rule:

$$x^{k+1} = D^{-1}(b - (L + U)x^k)$$

where D is the diagonal of the matrix A , L is the lower triangular part of A , and U is the upper triangular part of A .

Source:

- *A Multigrid Tutorial* by Briggs, et al. (2000)
- <https://yaningliucudenver.github.io/Numerical-Analysis-I/bookchapter3-2.html>

0.8 NGSolve Notes